Pranav Manjunath

469-534-1276 | pranav.manjunath@duke.edu | Personal Website | LinkedIn | Github

Rhodes Graduate Fellowship Awardee

PhD candidate passionate about transforming multi-modal data into insights, provide innovative solutions to complex real-world healthcare problems, build productive cross functional collaborations and lead by example.

Education

PhD in Biomedical Engineering — Aug 2022 – Present
Duke University (Advisor: Dr. Timothy Dunn) — *Durham, NC*

MS in Interdisciplinary Data Science (GPA: 3.95) — Aug 2020 – May 2022
Duke University — *Durham, NC*

BE in Computer Science and Engineering (Certificate of Distinction) — Aug 2016 – May 2020
Visvesvaraya Technological University (PESIT) — *Bangalore, India*

Experience

Member of Technical Staff, Intern (Research Scientist) — May 2026 – Present
Abridge — *New York City, NY*

- PhD Research Scientist Intern in the Strategic Research Team at Abridge.
- Reporting to the Head of Strategic Research at Abridge, in leading research to evaluate and measure the prevalence of shared decision-making and other care quality measures in real-world clinical conversations between patients and physicians.

Visiting Researcher — Oct 2025 – Present
Human-Computer Interaction Institute, Carnegie Mellon University — *Pittsburgh, PA*

- Collaborating with Dr. Adam Perer in the Data Interactions Group at CMU on evaluting and studying how multimodal AI systems can support clinical stakeholder and augment traumatic brain injury care.

Co-leader - Discovering AI Applications for Traumatic Brain Injury Care — Aug 2024 – May 2025
Duke University — *Durham, NC*

- Co-initiated and leading the Bass Connections project, securing over **$30,000 in competitive funding** to conduct the largest qualitative/mixed-methods research study focused on identifying AI-driven opportunities to improve TBI care at Duke. *Project Website*
- Collaborated with senior faculty across multiple departments to design and teach the project-based course to an interdisciplinary team of over 20 students.
- Awarded the **Rhodes Graduate Fellowship for Interdisciplinary Research** for spearheading this project
- Students present this research at the National Collegiate Research Conference 2026 at Harvard, Society-Centered AI Conference 2026, Duke National Health Research Conference 2026, and the Duke AI Health Data Science Showcase 2025.

Co-leader - Graduate School Committee: DukeREP — May 2023 – Present
Duke University — *Durham, NC*

- Outreach program aiming increase diversity in STEM fields by encouraging high-school students to pursue academic and professional careers in STEM
- Spearheaded efforts to recruit mentors and instructors within the PhD program to support and guide high school students through teaching and mentoring initiatives.

Health AI Researcher - +DS Program — May 2021 – May 2022
Duke University — *Durham, NC*

- Worked with Prof. Michael Pencina, Prof. Ricardo Henao and a team of Duke researchers on a NIH grant project to build a stroke survival predictive model.

Capstone Researcher - ViacomCBS/MTV — Aug 2021 – May 2022
Duke University — *Durham, NC*

- Worked with MTV to find solutions to increase the voter turnout rates amongst youth in the US elections using geospatial data. Tested a hypothesis that proximity of polling location to universities has an inverse correlation with the student voter turnout rate.

- Won **Silver Distinction at Anthem Awards 2023** under the category Awareness Media (for profit).
- Project Findings and Insights featured in *Business Wire* and *USA Today* - helping MTV to start social campaigns to improve student voter access.

Co President - Duke Interdisciplinary Product Management Club May 2021 – May 2022
Duke University *Durham, NC*

- Co-founder of a club at Duke to bridge Product Management, Data Science, and Machine Learning. Main responsibilities: Organize technical workshops, seminars, datathons, producthons, strategic case competitions

Project Manager - Data+ May 2021 – Aug 2021
Duke University *Durham, NC*

- Managed a team of Duke undergraduate students to develop machine learning features that can be used to identify unknown web attacks.
- Outcome incorporated into Duke's IT security infrastructure to help protect the network.

Data Scientist Summer Intern May 2021 – Aug 2021
Advance Auto Parts *Raleigh, NC*

- Clustering and topic modelling customer product reviews into key themes utilizing NLP unsupervised algorithms. Used by business leaders to address customer pain points and trends
- Feature engineered product, customer, location and vehicle data from various sources leveraging SQL and Python
- Presented results and recommendations to stakeholders, company technical interest group and C-suite executives.

Graduate Teaching Assistant Dec 2020 – Present
Duke University *Durham, NC*

- **Introduction to Data Science - Dr. Timothy Dunn**: Organize weekly office hours to review and teach course material and grade student assignments.
- **Design Health 3 - Dr. Eric Richardson, Dr. Paul Fearis, Dr. Joseph Knight**: Organize, create optimal class schedules templates and provide feedback and grade students on their presentations.
- **Programming for Data Analytics (Fuqua School of Business) - Prof. Brian Cozzi**: Organize weekly office hours to help students with coursework and teach them concepts in Python and R programming.
- **Data Engineering in the Cloud - Prof. Noah Gift**: Help structure the course, grade and review assignments, organize office hours to help students with the coursework.
- **Introduction to Machine Learning - Prof. Kyle Bradbury**: Help grade and review assignments, organize office hours to help students with the coursework.
- **Data Analytics and Applications (Fuqua School of Business) - Prof. Mattia Ciollaro**: Organize weekly office hours to help students with coursework and teach them concepts in Machine Learning.
- **Cloud Computing for Data Analysis - Prof. Noah Gift**: Help structure the course, grade and review assignments, organize office hours to help students with the coursework.
- **Python Winter Course and MIDS Bootcamp - Prof. Andrew Hilton and Prof. Genevieve Lipp**: Organize office hours for students to help teach them Python programming.

Publications

Temporal Real-World Evidence Retrieval for TBI Care Outcome Prediction with Language Models 2026

Pranav Manjunath, Brian Lerner, Timothy W. Dunn
In Review - ML4H Proceedings Track

How Much Information Is Enough? A Design Space for Mediating Personalized Information with AI in Emergency Care 2026

Pranav Manjunath, Tolu Oyesanya, Adam Perer, Timothy W. Dunn
ACM Conference on Human Factors in Computing Systems (CHI) 2026
Lightning Talk at CHI 2026 Workshop: Toward Relationship-Centered Care with AI

Trust Your Neighbors: Multimodal Patient Retrieval for TBI Prognosis 2025

Pranav Manjunath, Brian Lerner, Timothy Dunn
Journal: IEEE Journal of Biomedical and Health Informatics (JBHI)
Paper Link | Oral Presentation at IEEE BHI 2025 | **Best Paper Award**

Context-Aware Filtering of Unstructured Radiology Reports by Anatomical Region 2025
Zakk Heile, Pranav Manjunath, Brian Lerner, Samuel Berchuck, Monica Agrawal, Timothy W. Dunn
Conference: Machine Learning for Health (ML4H)
Paper Link

Personalized Case- and Evidence-Based TBI Prognosis with Small Language Models 2025
Pranav Manjunath, Syed M. Adil, Benjamin Wissel, Daniel Sexton, Brian Lerner, Timothy Dunn
Conference: IEEE Biomedical Health Informatics (IEEE BHI)
Paper Link | Oral Presentation at IEEE BHI 2025 | **Top 9% of Submissions**

Towards Interactive and Interpretable Image Retrieval-Based Diagnosis: Enhancing Brain Tumor Classification with LLM Explanations and Latent Structure Preservation 2024
Pranav Manjunath, Brian Lerner, Timothy Dunn
Conference: Artificial Intelligence in Medicine (AIME 2024)
Paper Link | Oral Presentation at AIME 2024 | **Best Student Paper Award**

Apriori Algorithm and Decision Tree Classification Methods to Mine Educational Data for Evaluating Graduate Admissions to U.S. Universities 2019
Pranav Manjunath, Kushal Naidu
Book Chapter: Algorithms for Intelligent Systems (Springer)
Paper Link

EyesPro: Protect Your Eyes 2018
Pranav Manjunath, Nimisha V. Arun
Journal: International Journal of Scientific Research in Science and Technology (RISE 2018 Inter-College Research Competition)
Paper Link | **Best Paper Award**

Posters & Abstracts

Collaborative Human–AI Systems for Supporting Information Needs in Acute Traumatic Brain Injury Care 2026
Pranav Manjunath, Timothy W. Dunn
Conference for Health, Inference, and Learning (CHIL) 2026
Doctoral Symposium

PORE-LM: A Personalized Outcome-aware Real-World Evidence Module for Language Models 2026
Pranav Manjunath, Brian Lerner, Bradley Kolls, Timothy W Dunn
AMIA Annual Symposium
Extended Abstract - Poster Session

Dual Retrieval for Personalized TBI Prognosis Using Real-World Evidence and Clinical Guidelines 2025
Pranav Manjunath, Syed Adil, Ben Wissel, Daniel Sexton, Brian Lerner, Timothy Dunn
Stanford AI+HEALTH 2025
Accepted Poster

Personalizing Total Knee Arthroplasty: A Deep Learning Approach to Restoring Natural Alignment with Anatomic Key-point Identification 2024
J. Helmkamp, Pranav Manjunath, B. Valan, P. Raghuvanshi, M. Izzi, Timothy Dunn, Seyler
Orthopedic Trauma Association (OTA 2024)
Accepted Abstract - Podium Presentation

AI-Driven Content-Based Image Retrieval to Support Brain Tumor Diagnosis 2023
Pranav Manjunath, Brian Lerner, Timothy Dunn
Conference: Duke AI Health Annual Conference
Accepted Poster

VNA to Z: An Automated Pipeline for Retrieving and Processing Duke Health CT Scans 2023

Brian Lerner, Pranav Manjunath, Timothy Dunn

Conference: Duke AI Health Annual Conference

Accepted Poster

Skills

Programming Languages: Python, R, SQL (MySQL and Postgres), C++

Cloud Services: AWS Machine Learning Certified, Google Cloud Platform

Extra Curricular Activities

Professional Violinist and Music Teacher

Trained in Indian Classical Carnatic Music

Co-founded Boston Brothers, an Indian classical and fusion violin band. Collaborated with international musicians, performed over 600 concerts worldwide, and have received multiple awards, honors and citations.

Secured 1st Rank in Vidwath, the highest Indian classical Carnatic music examination

Taught 25 students at a music academy in India and continues to conduct classes for several students in the US.

Contribute part of my earnings to charity, to support the education of poor students. I have performed for several fund-raising concerts to support local artists during the COVID-19 pandemic. As a band we have played in old age homes, charities, orphanages, to help spread the joy of music.